

Straight lines - Set 1 - Solutions

Specification MM3.1, M4.9, M4.10.

Answer key

Question	Answer	Topic
1	C	Straight lines
2	F	Straight lines
3	A	Straight lines
4	A	Straight lines
5	B	Straight lines
6	A	Straight lines
7	H	Straight lines
8	D	Straight lines
9	B	Straight lines
10	A	Straight lines
11	B	Straight lines
12	F	Straight lines

Worked solutions

Question 1 LIN-01 • Straight lines • spec MM3.1

Find the equation of the line through $(2, -3)$ that is perpendicular to $3x - 2y + 5 = 0$.

Answer: C

The given line has gradient $\frac{3}{2}$, so the perpendicular has gradient $-\frac{2}{3}$:
 $y + 3 = -\frac{2}{3}(x - 2) \Rightarrow 3y + 9 = -2x + 4 \Rightarrow 2x + 3y + 5 = 0$.

Question 2 LIN-02 • Straight lines • spec MM3.1

The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

Answer: F

Gradient $AB = \frac{2}{4} = \frac{1}{2}$. Gradient $AC = \frac{6}{k-1} = \frac{1}{2} \Rightarrow k - 1 = 12 \Rightarrow k = 13$.

Question 3 LIN-05 • Straight lines • spec MM3.1

Which of the following lines is parallel to $4x + 6y = 9$?

Answer: A

$4x + 6y = 9$ has gradient $-\frac{2}{3}$; so does $2x + 3y = 1$.

Question 4 LIN-06 • Straight lines • spec MM3.1

The perpendicular bisector of the line segment joining $(1, 5)$ and $(7, 1)$ has equation

Answer: A

Midpoint $(4, 3)$; gradient of segment $\frac{-4}{6} = -\frac{2}{3}$, so the bisector has gradient $\frac{3}{2}$:
 $y - 3 = \frac{3}{2}(x - 4) \Rightarrow 2y - 6 = 3x - 12 \Rightarrow 3x - 2y = 6$.

Question 5 LIN-07 • Straight lines • spec MM3.1

The line $ax + by = 12$ has x -intercept 3 and y -intercept -4 . Find $a - b$.

Answer: B

$(3, 0)$: $3a = 12 \Rightarrow a = 4$. $(0, -4)$: $-4b = 12 \Rightarrow b = -3$. $a - b = 7$.

Question 6 LIN-14 • Straight lines • spec MM3.1

A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .

Answer: A

Midpoint $(1, 2)$ means $A = (2, 0)$ and $B = (0, 4)$. Gradient $= \frac{4-0}{0-2} = -2$.

Question 7 X2-05 · Straight lines · spec MM3.1

Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

- I If $m_1m_2 = -1$ then the lines meet at exactly one point.
- II If $m_1 = m_2$ then the lines do not meet.
- III If the lines meet on the y -axis then $c_1 = c_2$.

Answer: H

I: perpendicular lines are not parallel, so they meet once. II: equal gradients and distinct lines means parallel and different, so no intersection. III: on the y -axis $x = 0$, so $y = c_1 = c_2$.

Question 8 LIN-03 · Straight lines · spec MM3.1

The line L passes through $(4, 0)$ and $(0, 6)$. Find the perpendicular distance from the origin to L .

Answer: D

Area of triangle $= \frac{1}{2} \cdot 4 \cdot 6 = 12$. Hypotenuse $= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$. Distance d satisfies $\frac{1}{2} \cdot 2\sqrt{13} \cdot d = 12$, so $d = \frac{12}{\sqrt{13}}$.

Question 9 LIN-04 · Straight lines · spec MM3.1

The lines $y = 2x + 1$, $y = -x + 7$ and the x -axis enclose a triangle. Find its area.

Answer: B

Intersection: $2x + 1 = 7 - x \Rightarrow x = 2$, $y = 5$. x -intercepts: $-\frac{1}{2}$ and 7 , base $= \frac{15}{2}$. Area $= \frac{1}{2} \cdot \frac{15}{2} \cdot 5 = \frac{75}{4}$.

Question 10 LIN-08 · Straight lines · spec MM3.1

For which value of k are the lines $kx + 3y = 2$ and $2x - (k - 1)y = 5$ perpendicular?

Answer: A

Gradients $-\frac{k}{3}$ and $\frac{2}{k-1}$. Perpendicular when the product is -1 : $\frac{-2k}{3(k-1)} = -1 \Rightarrow 2k = 3k - 3 \Rightarrow k = 3$.

Question 11 LIN-09 · Straight lines · spec MM3.1

The point P lies on the line $y = 2x - 1$ and is equidistant from $(0, 0)$ and $(4, 2)$. Find the coordinates of P .

Answer: B

Equidistant means P is on the perpendicular bisector of $(0, 0)$ – $(4, 2)$: midpoint $(2, 1)$, gradient -2 : $y = -2x + 5$.
With $y = 2x - 1$: $4x = 6$, $x = \frac{3}{2}$, $y = 2$. So $P = (\frac{3}{2}, 2)$.

Question 12 LIN-11 · Straight lines · spec MM3.1

The vertices of a triangle are $A(0, 0)$, $B(6, 0)$ and $C(2, 4)$. Find the equation of the line through C that bisects the area of the triangle.

Answer: F

A line through a vertex bisects the area iff it passes through the midpoint of the opposite side, $(3, 0)$. Through $(2, 4)$ and $(3, 0)$: gradient -4 , $y = -4x + 12$.